Density Related Supply Side Regulatory Constraints and Housing Affordability in American Cities

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HONOR PLEDGE:

This paper represents my own work in accordance with University regulations.

Signature

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Introduction:

The United States is in the midst of a housing affordability crisis of monumental proportions. In the past few decades, housing has mutated from a mundane task of bureaucratic city planning, to a central policy concern in many major cities across the US. Present relative housing prices are historically quite high, and depending upon how one defines affordability metrics, they are at all-time historical highs (Longtermtrends 2026; Whitney 2025). Median home prices have risen faster than incomes have over the past decade. In turn, rental markets have also followed. Median rents now far eclipse personal incomes in many major cities across the country, compressing personal finances and forcing workers to consider alternative locations to live.

Economists largely default to a supply-side explanation of why housing prices continue to skyrocket. They attribute these increases in housing costs to a lack of housing inventory, and the intuitive ensuing logic is that the freeing up of housing inventory, primarily in the form of the construction of new housing stock, will drive down prices and rents across the market. The intuition is well grounded in classical economic theory: when supply is artificially constrained and inelastic, prices will rise above what the underlying marginal costs of production would otherwise dictate. Glaeser and Gyourko (2003) pertinently named this phenomenon the regulatory "zoning tax", defined as the markup of market prices over construction costs, not attributable to land scarcity or labor costs, but rather to the legal constraints imposed by local governments on what can be built where, and how much of it can be built. Orthodox wisdom has it that regulatory barriers limit the supply of housing relative to demand in certain

high-demand, inelastic markets, driving up housing prices. In this manner, it is argued that the proliferation of excessive local regulatory regimes is a key contributor to the lack of housing affordability observed in the United States today.

The primary question at hand that our research aims to explore is the manner in which the presence of a restrictive regulatory regime with respect to the zoning & permitting of new housing construction influences housing affordability in cities in the United States. Further, several secondary questions that we intend to answer are whether or not stricter land use regulations correlate with higher median home prices.

Empirical Motivations:

Glaeser and Gyourko have published several seminal papers examining the relationship between regulatory restrictions on new housing stock and its consequences for housing affordability (2002 & 2003). They found that both building restrictions, as well as zoning restrictions, had a significant and measurable negative impact on their candidate measure of housing affordability, defined by housing price per square foot.

Similar papers published in the ensuing years have echoed these results. A 2005 paper found a relationship between regulatory stringency and housing prices in California (Quigley & Raphael). A 2007 paper by Keith Ihlanfeldt examining over 100 cities in Florida found that greater regulatory stringency increased housing prices (2007). A 2011 paper found that minimum lot sizes, specifically, were associated with significantly higher housing prices in Massachusetts (Zabel and Dalton). A 2018 study found that the cost share of housing regulatory constraints specifically, not land costs, or

construction, to be responsible for the supply-side share of rising housing prices (Albouy & Ehrlich, 2018)

Other studies, conducted internationally, add to the growing litany of evidence that housing prices and housing affordability lie inherently downstream of the regulatory constraint on the housing supply (Hilber & Vermuelen 2015). Moreover, Hilber & Vermuelen found that stricter land use constraints on housing in England significantly increased the sensitivity of housing prices to income growth, providing a clear mechanism for skyrocketing housing prices in high-demand municipalities.

In a 2022 article published in the Journal of Urban Economics, Molloy and colleagues also found that increased housing supply constraints, in the form of increased regulation limiting the construction of new housing, were predictive of increased housing prices, even when adjusted for housing quality. And in their more novel finding, they further concluded that demand for housing type was much less elastic than one might naively assume. That is, householders generally do not adjust their preferences for unit or lot size in the face of increasing housing prices. The inelasticity of demand with respect to housing implies that if households do not readily substitute towards smaller or denser units when prices rise, then regulations that restrict the strata of housing types available to consumers on the market will put further pressure on housing prices for preferred strata, thus reducing home affordability.

This paper addresses the gaps left in the literature. Our primary contribution is a decomposition of the effects of different regulatory mechanisms on housing affordability in US metropolitan areas, instead of regulation in the aggregate. Here, we employ three

central sub-indices of the Wharton Land Use Regulatory Index (WRLURI), invented by our familiar friends Glaser and Gyourko: the Density Restrictions Index (DRI) which measures the stringency of minimum lot size requirements that constrain allowable build density, the Supply Restrictions Index (SRI), which reflects the extent of explicit annual caps on permits and construction; and the Approval Delay Index (ADI), which measures the average duration of the permitting and approval process, which is one of, if not the largest hurdle facing developers who try to bring new houses to market. We examine these instruments simultaneously, as well as independently, in a simple multivariate cross-sectional OLS regression model with state fixed effects, aggregated across Core-Based Statistical Areas (CBSAs) defined by the Census Bureau with data derived from the annual American Community Survey. We are able to estimate whether or not regulation affects housing affordability in the aggregate, but also which mechanisms have the most pronounced impact, and the heterogeneity in impact across these dimensions, if it exists. This is our primary contribution to the literature.

Our secondary contribution is extending the analysis. We move beyond simple hedonic regression upon nominal housing prices as the primary dependent variable of interest by exploring alternate measures of affordability. We construct normalized price to income and rent-to-income ratios that are more directly comparable across metropolitan areas of varying size, income, and demand intensity, and likely also more independent of the local price level. This allows us to examine whether or not the regulatory premium, Glaeser's "zoning tax" that exists on top housing costs represents a true diminution of purchasing power, or whether or not it is at least partially absorbed by higher local incomes.

We posit that the existence of greater regulatory restrictiveness in local jurisdiction's land use regulation, specifically pertaining to the extent that they restrict the available supply of housing in the jurisdiction, exerts statistically meaningful upward pressure on housing prices and decreases housing affordability, as measured by the nominal prices and the price to income ratio. We employ the use of the Density Restriction Index of the WRLURI, the Supply Restrictions Index and the Approval Delay Index of the Wharton Residential Land Use Regulatory Index to estimate how these local jurisdictions restrict the available supply of housing to a greater degree of granularity, and estimate which dimensions of regulation are the most impactful and restrictive.

Methodology:

For this thesis, we employ a quantitative, quasi-experimental research design in order to thoroughly investigate the relationship between local regulatory land-use restrictiveness and housing affordability in the US across major metropolitan areas.

In order to investigate our primary research question, how restrictive local zoning regimes impact housing affordability, we design our empirical model to employ a multivariate cross-sectional Ordinary Least Squares (OLS) regression with supplementary panel data components and conduct an in-depth analysis of our results. We exploit the extensive geographic coverage and local nature of the WRLURI to explore our question at the level of CBSAs. By regressing housing costs, or cost instruments, on the WRLURI, we are able to isolate the relationship between specific types of land use regulation and their effects on housing affordability.

The Empirical Model

The full model specification takes the following form:

$$Affordability_i = \alpha + \beta_1 * WRLURI_i + \sum_{j=2}^{k} \beta_j Control_{j,i} + FE_i + \epsilon_i \text{ and}$$

$$Affordability_i = \alpha + \sum_{j=1}^{k} \beta_j * WRLURISUB_{j,i} + \sum_{j=k+1}^{n} \beta_j Control_{j,i} + FE_i + \epsilon_i$$

Where Affordability is one of our candidate outcome measures, defined here as median home value or price-to-income ratio for a CBSA ***i*** in the ACS survey year (either 2006 or 2018); WRLURI is the composite Wharton Residential Land Use Regulatory Index score for that CBSA; WRLURISUB represents the array of WRLURI subindices, where j denotes the chosen sub index--specifically, we analyse the Density Restrictions Index, Supply Restrictions Index, and the Approval Delay Index as our candidate sub-indices of analysis–included specifically to disaggregate any observed regulatory effect into its constituent mechanisms; Control j,i represents our vector of demographic, economic, and supply side controls; FEi represents the state level fixed effects term, and ϵ_i is the robust error term, clustered at the CBSA level to account for within-metro correlation across the ACS.

We build out the model in our analysis piece by piece, starting with a barebones, raw bivariate regression model that regresses each affordability outcome on the composite WRLURI score alone, establishing the baseline relationship. We then add our control variables progressively, starting with supply characteristics in order to isolate the contribution of regulatory tightening to the supply availability specifically, then

demographic and economic controls to control for the demand side contribution to housing prices, then state fixed effects. We do this in order to observe how the WRLURI coefficient is attenuated (or not) as confounding variables are absorbed. Finally, we replace the composite WRLURI with the independent, as well as simultaneous, inclusion of the DRI, SRI, and ADI sub-indices, with controls and fixed effects, in order to assess which regulatory mechanisms drive the aggregate result.

Empirical Model: Supplementary Concerns.

We believe that OLS is the most appropriate estimator for this analysis for several reasons. Number one, our outcome variables are all continuous and approximately normally distributed across CBSAs–our primary level of analysis–making assumptions of linearity reasonable. Furthermore, the WRLURI and its sub-indices are all normalized and continuous, and we theoretically expect the relationship between regulatory stringency and housing costs to be approximately linear, and monotonically increasing within the range of observation. We do not have strong priors that the effect is nonlinear, but we do test for this in order to ensure that this is not the case. We ensure that the standard errors are robust throughout our analysis, because we do reasonably expect heteroskedasticity in several variables of interest, primarily, but not limited to income and housing costs.

We define the CBSA as the fundamental unit of observation, because we believe that this level of analysis captures the most economically meaningful unit of housing market activity. We believe that housing markets fundamentally lie downstream of labor markets. People desire to live where jobs are present. Thus, local housing markets are

not strictly defined by municipal borders, but more so by commuting zones and labor market aggregation areas that cast an economic sphere of influence that determines where households may reasonably choose to live in order to access the jobs in that particular market.

The CBSA is ideally situated to capture this relationship. CBSAs are drawn up by the Office of Management and Budget (OMB) on the basis of commuting patterns between an urban core and its surrounding neighborhoods, chosen directly because of the degree of economic and social incorporation between the two. Someone who opts to live in Morris County New Jersey, for example, dreading the non-stop hustle and bustle of New York City, is still operating within the same broader market and commuting zone, even though Morris County is located roughly 30 miles away from Manhattan, and even though those two counties fall into completely different legal jurisdictions–even different states.

Analyzing the effects of regulatory stringency at a more granular level, such as the county level, would impose artificial boundaries that may be too restrictive in many larger, integrated markets, and analyzing it at the most granular level, the Census Place or Minor Civil Division (MCD) level would either gloss over economic integration entirely or a more rigorous approach would necessitate estimation of cross-jurisdictional regulatory price spillovers–that we are not really adequately prepared to investigate or model.

There is an argument to be made that despite this common economic tie, the vast majority of the difference in prices does indeed lie in the regulatory effect, and if we

are interested in generating the most statistical power in order to isolate the regulatory effect, then we shouldn't concern ourselves so much with controlling for the demand side, and instead seek to analyse the effect of regulatory restrictiveness with the greatest degree of granular control.

The choice to perform our analysis at the CBSA level is not without cost. As we discuss further on in our analysis section, aggregating place-level WRLURI scores at the CBSA level necessarily blurs over intra-metro regulatory heterogeneity. There may be some metropolitan areas that are characterized as having highly restrictive suburbs and permissive core cities, or vice versa, so the global average WRLURI score for a given MSA may not reflect with an adequate degree of granularity the binding regulatory constraints experienced by some communities lying within the boundaries of the MSA. This blurring is a potential drawback that could severely limit the statistical power of our analysis.

If the binding constraint on housing supply within a metro is concentrated in a handful of suburbs with extreme DRI scores, or a dense urban core with extreme ADI scores, averaging those scores with the more permissive jurisdictions in the level of analysis will depreciate our coefficients' estimates and cause them trend towards the global mean. This phenomenon, quite clearly, reduces the odds that we will find a statistically significant effect. On the other hand, if such statistically significant coefficients are found, it is all the more reason to take them seriously, as, despite a dearth of object-level granularity, our measurements have been able to educe a statistically significant result – even encumbered by the data abstruseness consequent to a blurry estimand. Our estimates should be interpreted as conservative lower bounds

on the true effect of regulatory restrictions, at least to the extent that intra-metropolitan heterogeneity is real.

Data Sources and Sample

The WRLURI.

The primary independent variable in our analysis, and a direct instrument for measuring the stringency of the local housing regulatory environment with respect to zoning & permitting, is the 2018 Wharton Residential Land Use Index. Functionally, the WRLURI is a standardized metric of land-use restrictiveness, estimated simultaneously across multiple dimensions and constructed from a comprehensive survey of around 2500 jurisdictions across the United States. It can be fractionated into 12 separate sub-indices, each of which is intended to appropriately capture some dimension, or mechanism, of the regulatory environment.

The sub-indices are created by selecting several individual survey questions related to that sub-index, as well as, in some cases, combining additional determinative statistical factors. These individual factors are mathematically interpreted, aggregated, and manipulated in accordance with an individualized points-based coding schema assigned to each of the questions. Factor analysis was employed to derive a unique weighting for each constituent factor in order to produce the 12 continuous sub-indices. The sub-indices are combined together via factor weighting and then statistically normalized to achieve a weighted aggregate index of the land use regulatory environment in that community.

The WRLURI sub-indices of choice that we specify for sub-index decomposition analysis in this study are: the Density Restrictions Index (DRI), Approval Delay Index (ADI), and the Supply Restrictions Index (SRI).

The Density Restrictions Index is a simple, continuous and monotonic constructed variable that is intended to capture the magnitude of density restrictions that exist, specifically pertaining to minimum lot sizes. From Gyourko et al: The DRI subindex takes on values ranging from 0 to 4 and is constructed as follows:

DRI = 0 if there is no minimum lot size regulation anywhere in the jurisdiction

= 1 if there exists a minimum, no larger than 0.5 acres

= 2 if there exists a minimum, and the largest is between 0.5-1 acres

= 3 if there exists a minimum, and the largest is between 1-2 acres

= 4 if there exists a minimum, and the largest is greater than 2 acres.

The Supply Restrictions Index "reflects the extent to which there are explicit annual caps on the supply of new housing". It is defined as the simple sum of the number of limits on building permits, construction, number of dwellings or units, and is also continuous and monotonic. It is constructed in the following manner, with possible values ranging from 0 to 6 (Gyourko 2021):

SRI = SFPermits + MFpermits + SFConst + MFConst + MFBuild + MFUnitsDwell

The Approval Delay Index (ADI) is the final subindex contributing to the aggregate WRLURI and measures permit approval timelines. The index combines

average review times for residential projects, multi-unit rezoning applications, and subdivision requests. Gyourko et al. define the canonical construction of the ADI as such:

> The ADI is calculated in several steps. First, we compute the simple averages of review time (reported in months) for:
>
> (1) by-right single-family unit projects and by-right multifamily unit projects
>
> ((sfprojrev + mfprojrev)/2) from Questions 16a and 16b);
>
> (2) not by-right single-family unit projects and not by-right multifamily unit projects ((nsfprojrev + nmfprojrev)/2) from Questions 17a and 17b);
>
> (3) application for and issuance of a building permit for development of less than 50 single-family units, 50 or more single-family units, and multi-family units
>
> ((sfl50 + sfm50+ mf)/3) from Questions 20a, 20b, and 20c);
>
> (4) application for subdivision approval and issuance of a building permit for development of less than 50 single-family units, 50 or more single-family units, and multi-family units ((subsfl50 + subsfm50 + submf)/3) from Questions 22a, 22b, and 22c).
>
> We then take the average of these four numbers to arrive at the number of months involved in the Approval Delay Index (ADI).

The American Community Survey.

The primary dependent variable in our analysis is housing affordability. Here, we construct and examine several candidate measures to instrument housing affordability in order to ensure adequate robustness and thoroughness in our analysis.

We extract the following variables from public use microdata series data taken from the yearly American Community Survey (ACS) , accessed via the IPUMS-USA website (ipums.org). We construct our variables from the 1-year ACS estimates at the level of the metropolitan statistical area. Our primary ACS-derived variables are: Home value, derived from the valueh variable in the ACS data; Price to Income Ratio: constructed by dividing the home value variable by the log transformed household income variable, providing an instrumented measure of affordability independent of the nominal price.

Secondary Analysis.

In order to properly address and dismiss concerns about the potential for reverse causality in the cross-sectional study design, whereby communities that are already expensive may adopt stricter land use regulations, either in response to rising prices or in an effort to gatekeep the community from outsiders, we conceived of a hypothetical a secondary differences-in-differences (DiD) study design, wherein we would exploit the temporal variation in between the original 2006 WRLURI survey and the updated 2018 WRLURI survey wave that is central to our cross sectional analysis. This specification would have compared the differences in housing affordability outcomes between a treated group, consisting of metropolitan areas that experienced the greatest increases

in regulatory stringency between the two waves of the survey, 2006 and 2018, and an untreated control group, those that experienced the least increase, or potentially even a decrease in regulatory restrictiveness over the same period.

The underlying assumption is that in the absence of deliberately pursued differential patterns of land-use regulation across the two groups, treated and control metros would have otherwise followed the similar trends in housing prices, or affordability metrics. This assumption is at least somewhat supported by the pre-period balance checks that we performed, which thoroughly rejected the idea of divergent pre-treatment trends. These balance checks confirmed that metros in the top and bottom terciles of regulatory tightening showed no statistically significant differences in housing affordability, income, density, employment, or population growth in the pre-treatment period.

Unfortunately, however, the particular intended implementation of the DiD study design angle was precluded by a geographic unit incongruencies between the 2006 WRLURI datafile and the 2018 WRLURI datafile. The 2018 WRLURI dataset identifies communities by Census place Federal Information Processing System (FIPS) codes as its base level of specification – however, not all of the place FIPS observations in the 2006 datafile correspond to actual place FIPs code, but rather to the frustrating contrary, a substantial minority of the observations (around 500) instead correspond to MCD FIPS codes which needed to be crosswalked separately, and caused econometric headaches while performing multiple many onto many merges in quick succession.

The 2006 WRLURI dataset also identifies communities by Census place FIPS codes as its base level of specification, but the inability to procure a 1999 place to county crosswalk in sufficient time forced us to specify the 1999 OMB Metropolitan Statistical Area (MSA) codes as our level of delineation, making it impossible to harmonize the 2006 and 2018 WRLURI at a common unit of observation without substantial loss of granularity or statistical power. If such a crosswalk were obtained, it may be possible to perform the analysis, however, one would still have to crosswalk the 1999 file to the 2000s census assignments, then crosswalk the product of the resulting merge to the 2010 assignments via a 2000 to 2010 assignment crosswalk, and subsequently the 2010 assignments to the 2013/2018 assignments, manually correcting any assignment errors arising from the post-census code reassignment shuffles at each step before continuing. In short – they have their work cut out for them.

In light of these unfortunate econometric headaches, we therefore present our cross-sectional results as our primary analysis and treat the DiD specification as a brief footnote and a direction for future research. We do, however, briefly present the neutered DiD specification performed at the MSA level in our findings as a piece of supportive, tertiary evidence for our theories–noted with the caveat that it is not a particularly powered piece of evidence.

Further Methodological Considerations:

One natural approach to teasing out endogeneity in regulatory restrictiveness, if it exists, particularly with regards to the density restriction component of the WRLURI, is to echo Saiz (2010) in using geographic instruments for land availability as a source of

exogenous variation in housing supply constraints. In the paper, Saiz demonstrates that physically constrained metropolitan areas, determined by measuring the share of land within 50 kilometers of the city center that is rendered unbuildable by the presence of bodies of water or steep terrain (grade above 15%), exhibit significantly lower housing supply elasticities and higher price responses to demand shocks than geographically unconstrained metros. Saiz further employs the WRLURI to tease out the regulatory contribution to housing supply elasticities and price response to demand shocks. The intuition is simple and to the point: a city that is physically geographically constrained by an ocean, bay, or mountains, simply cannot expand its housing stock topographically outward–in a manner that is typical for American sprawl-centered development–in response to significant demand pressure, regardless of the regulatory environment in the city. Thus, geography is a plausibly exogenous determinant of, or contributor to, the effective supply constraint in the city that is independent regulation.

We explored the feasibility of incorporating some Saiz-like land (un)availability measures as an approximate control variable in our baseline housing price regressions in order to cleave the contribution of the effects of physical constraints on supply from the contribution derived from legal/regulatory constraints on housing supply. In other words, we are interested in the marginal contribution of regulatory tightness to determining housing prices, conditional on the physical environment. Ideally, one would try to hold the geographical constraints constant and estimate the residual effect of regulation, rather than use geography as an instrument for regulation itself in the manner that Saiz did. However we ultimately concluded that the approach faced both

practical and conceptual obstacles that made it not worthy of consideration for inclusion in our analysis.

Primarily, the Saiz land instrument is simply not very compelling as a standalone constraint on housing availability, as developers can always build upward – so long as they are legally permitted to. This means it cannot truly be inserted into our regression as a standalone control. Any proper integration of a Saiz-like instrument should include an interaction term with the regulatory restrictiveness variable, as the two most likely display a dyadic, non-linear and mutually intensifying relationship–that is, legal/regulatory supply constraints intensify the effects of geographic supply constraints, and vice versa. This obviously introduces additional layers of complexity into the model, which may place it thoroughly beyond the scope of a senior thesis investigation.

Moreover, as a specification, it does not capture the full extent of geographic constraints on housing supply in practice. It can be reasoned that, especially in concert with the legal/regulatory dimension, there are in fact many more geographical constraints that prevent builders from supplying new houses than just mountainous terrain and water. State, federal, or locally owned government land, environmentally protected areas, flood, hurricane, wildfire or otherwise high risk weather or natural disaster zones, other forms of land that are simply impractical to build on, such as clay, silt, or peaty soil, swampy lands, plains prone to sinkholes, parcel irregularities, inadequate infrastructure access, or even water scarcity are all factors that may exert a chilling effect on supply availability, simply by making it infeasible to erect new construction.

In light of this, we explored an alternative approach to controlling for physical supply constraints, spurred on by a similar hypothesis. We considered that metropolitan areas that are already densely populated and experienced most development prior to the advent of modern zoning (nearly every big city in the US) in practice have endogenous physical constraints on new construction merely as a consequence of the pre-existing density itself. If a city was built out to high density before zoning ordinances were adopted (here, we would set a cutoff at the end of WWII, in light of the propagation of modern zoning laws in the postwar era) it in practice faces a dramatically different development constraint than a city that was sparsely populated before it adopted modern zoning laws.

Regardless of the degree of *de jure* regulatory constraint that developers are subjected to in these two cities, the more sparsely populated city of the two will always afford more de-facto freedom to develop. The more dense cities are constrained in reality by existing built infrastructure, legacy lot subdivisions, economic constraints on land availability, demolition, and redevelopment, and the interaction of the zoning code with these physical constraints. The reasoning goes that cities that experienced much of their development after the advent of zoning presumably have much more consistent lot divisions, infrastructure access, and coordinated land use patterns in general, which makes it easier to erect new construction that is in line with the existing regulatory constraints. Present day legal compliance is more clean cut and requires much less time navigating the labyrinth of variances, conditional uses, and easements.

We considered controlling for pre-war population density, prior to the widespread adoption of exclusionary zoning, as a proxy for this built-environment constraint. Given

the constraints of the Census and ACS data that we are working with, we would have established a cutoff around 1940 or 1945 (interpolating the data from 1940 and 1950), using county-level data aggregated to the MSA/CBSA level. And while we considered the approach to be highly conceptually appealing, we ultimately did not include this pre-war-density manufactured control as a control in our regression specifications due unresolved implementation complexities, data availability constraints, and the lack of a sufficiently robust theoretical justification.

Instead, we note it here as a theoretically motivated control that would have solidified the robustness of our causal interpretation of our analysis, and we proffer that it represents a natural extension of this thesis. Therefore, our primary identification strategy was simplified to relying on the inclusion of natural demand side controls and state fixed effects to reduce residual confounding, and few simple endogeneity tests to strengthen our certitude in the strength of our analysis. We fully acknowledge that residual endogeneity cannot be fully eliminated in our cross-sectional framework, however, we are very confident in the statistical power that we harnessed to compute the regression specifications that we did ultimately include in our primary strategy, and especially confident in the statistical significance of the results that we did achieve.

Combining the Datasets.

As the WRLURI is determined at the census place level, while the highest degree of granularity in the ACS integrated public use microdata is only at the metropolitan area level, we were required to devise a mapping in order to equivalize the granularity of the dataset for our analysis.

In order to harmonize the heterogeneous geographic units, we made use of National Bureau of Economic Research (NBER) Census Core-Based Statistical Area (CBSA) to Federal Information Processing Series (FIPS) County Crosswalk files. We systematically aggregated each WRLURI community and mapped it onto its corresponding CBSA from the ACS, and then collapsed the community-level regulatory measures into a single, representative value for each CBSA. This approach enables a many-to-one merge in which all annual observations for a given MSA in the ACS panel are merged onto a single time-invariant WRLURI score. We were able to further operationalize this approach and individually merge our sub-indices of analysis onto the dataset and also merge values from the original 2006 WRLURI onto the dataset as well, treated as a separate variable.

The first step was achieved by linking each 2018 FIPS place code identified WRLURI community to a corresponding CBSA code. As there are no direct Census Place to CBSA crosswalks that we have access to, we employed an indirect workaround. We matched place FIPs codes from the WRLURI to their mother counties, also included in the WRLURI dataset, then merged the WRLURI dataset to the NBER CBSA to FIPS county crosswalk files according to county FIPs codes (NBER 2026). Thus, every place corresponding to a FIPs code was assigned a county FIPS code, and every place that was located within a CBSA--metropolitan or micropolitan statistical area--was assigned a CBSA identifier. Those that did not, are excluded from the analysis, as it focuses on metropolitan communities. Then, we collapsed this composite dataset, as mentioned above, to an array consisting of WRLURI candidate sub-indices of analysis (the composite index and the candidate sub-indices) separately, by each

CBSA code and then merged these variables m:1 with the ACS data according to the CBSA identified.

For this merge, we employed the use of the 2020 CBSA delineations. Because the 2018 CBSA delineations were created specifically to preliminarily conduct the 2020 census, the two should in theory be nearly, if not completely, geographically identical. There were a small number of observations (less than 1% of the total observations) that did not successfully match during the merge. We were not able to detect any issues in our preliminary regressions as a result of this strategy and it should not detrimentally impact the granularity of our analysis to a highly significant degree.

The simple reasoning behind the choice of employing CBSAs as our primary level of specification and analysis is because we hypothesize that CBSAs are more representative of the local political and economic environment of a geographic area, arising from the sphere of influence of a given city, or group of cities, than the county level is. That is, large, metropolitan area defining cities like New York City, for example, likely exert greater economic and political influence ("ripple effects") over neighboring towns and cities than county borders and governments do.

For the 2006 WRLURI, the merge was substantially more complicated and involved more steps. The 2006 WRLURI file contained no usable Census Place FIPS codes or county FIPS codes per-se. What it did contain were two geographic identifier variables: internal survey ID, and a 1999 MSA code identifier variable msa99. The first step was achieved by linking each of the 1999 FIPS MSA codes in the 2006 WRLURI file to a corresponding county level FIPS code for the 2000 census, utilizing GEOCORR

crosswalks. Thus, every place corresponding to an MSA was assigned a county FIPS code. Because the 2000 Geocorr file used PMSA level codes for large metropolitan areas, while the WRLURI msa99 variable used CMSA level codes, we had to manually recode many of the observations in order to successfully perform the merge.

Once merged, we once again had to crosswalk the 2000 counties to 2010 counties, merge the datasets, and then manually fix any errors, and then crosswalk the 2010 counties to the 2020, or 2013 definitions, then merge and manually fix any errors. It was a very laborious and time consuming task, however, it was accomplished successfully. Finally, the 2006 WRLURI file was merged into the ACS dataset via the 2013 MSA identifier variable.

Unit Quality Controls:

Adding bedrooms as a quality control that accounts for variation in the size of the house, we indeed see that bedrooms are negatively correlated with housing prices, and absorb some of the raw correlation between the WRLURI and home prices. While this relationship is highly counterintuitive on its face, it is actually a well-observed finding observed with reasonable frequency in hedonic housing price regressions that increasing the number of rooms or bedrooms, when controlling for the unit size, does not increase the value of the property, and in some instances, may counterintuitively decrease the value of the home (Rasmussen & Zuehlke 1986). This is likely because increasing the number of bedrooms without increasing the square footage of the home reduces the quality of the home.

Although we did not control for square footage in this naive regression, we still nonetheless observed this relationship in our data. We postulate that this is likely due to the fact that many older buildings, built in the immediate post-war era, have much lower square footage than modern homes, but proportionally much more rooms compared to the square footage. And further, these older post-war era homes represent lower quality housing stock that command substantially reduced prices in the marketplace.

We are reasonably confident that this phenomenon, having been documented prior, will not affect the robustness of our analysis in the grand scheme of things. Nevertheless, in order to control for these discordant results, we explored several alternative approaches to controlling for unit quality and size.

We first tried defining a room to bedroom ratio variable and instrumenting it as a proxy for home quality and use of space. A bathroom to bedroom variable might be more apt, as the bathroom to bedroom ratio has been found previously in the literature to be significantly more predictive of housing prices than bedrooms or bathrooms alone or in tandem. However, a critical and unfortunate limitation of the American Community Survey dataset that we utilized is that it has not reported a bathroom variable in the dwelling characteristic data since 1990–far too early to be useful for our analysis. Thus, we tried to instrument a rooms to bedrooms ratio variable in lieu of this. The logic follows that houses that have many bedrooms and not much living space will have much lower ratios than homes that have many bedrooms, but also much living space, or few bedrooms and much living space.

The intuition is that houses that possess the characteristic of having many bedrooms relative to total rooms are likely to be older stock, with smaller individual rooms, while homes with high room-to-bedroom ratios have more living space relative to sleeping quarters. Additionally, more modern homes and more expensive homes tend to have many more rooms on average than older stock, built for function on simplistic floorplans. However, the coefficient on the room to bedroom ratio remained negative, likely because it captures the urban-suburban split rather than unit quality per se. Urban homes tend to be more expensive, yet have fewer bedrooms and lower room-to-bedroom ratios than suburban homes of lower value. Substituting total rooms for the ratio yielded similar results for the same reason.

We tried including the decade that the dwelling was built as a vintage – captured through an array of dummy variables constructed from the builtyr2 variable – and the unit structure type as additional controls. Structure type is particularly important, as we postulate that the hedonic relationship between dwelling characteristics and housing price is likely highly heterogenous across different structure types – ie. single family homes display different feature to price relationships than single family attached units or multi-family buildings. We included the age of the structure as a control, as older homes, such as those built during the post war housing boom, are more likely to have proportionally more bedrooms with proportionally less overall living space. However, after all was said and done, the coefficient on room to bedroom ratio was still negative, likely due to capturing differences in home values across the suburban/urban split.

At this point, in lieu of further exploring the issue, given the complexity of the square footage issue and particularly its limited impact on the primary coefficient of

interest: the WRLURI, we elected to proceed with total rooms, structure type dummies, and a control specification for structure age consisting of an array of dummy variables for the decade corresponding to the structure's erection as our unit quality control vectors. We acknowledge that the absence of direct square footage data and number of bathrooms per structure in the ACS microdata remains a minor limitation of our analysis.

Results:

Main Regressions:

To begin our analysis, we define and run the first raw bivariate regression by regressing the raw variable corresponding to house value in the ACS survey, valueh, on the aggregate WRLURI index. We opt to run this regression with robust standard errors, as the default assumption of homoskedasticity is likely unsupported. It seems reasonable that the variance in home values, as well as that of future, yet to be added control variables, may be altered as we increase or decrease the value of the WRLURI.

Table 1: OLS Estimates: Home Value

	(1)	(2)	(3)
VARIABLES	Baseline	Controls	Controls + State FE
WRLURI (2018)	206,938***	139,816***	68,614***
	(161.7)	(203.5)	(253.3)
Population Density		6.648***	6.210***
		(0.0338)	(0.0354)
Household Income		1.531***	1.459***
		(0.00241)	(0.00238)
College Attainment Rate		73,649***	69,961***
		(234.6)	(227.6)
Employment Rate		-1.314e+06***	4.009e+06***

		(9,429)	(15,379)
Year Built		5,728***	
		(37.51)	
Population Growth Rate		1,255	-86,918***
		(885.5)	(1,320)
Constant	326,106***	1.278e+06***	-3.703e+06***
	(71.65)	(8,830)	(14,380)
Observations	30,482,342	15,002,342	15,002,342
R-squared	0.060	0.258	0.300
R-squared	0.0603	0.258	0.300

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

We observe a consistent and strong positive correlation between the WRLURI aggregate index and raw home value. For every 1 point increase in the WRLURI, that is, a 1 standard deviation increase in regulatory restrictiveness as measured by the WRLURI index, the predicted value of a home increases by $206,938. At first glance, this is a very sizable and healthy correlation, so it seems almost certain that a relationship between the two variables is assured. However, this robust association is likely largely being propped up by demand side variation. Metropolitan areas that score high on the WRLURI index are empirically observed to be more likely to be high-income, high-amenity, high-demand areas, independent of their regulatory environments. Cities like e.g. New York, San Francisco and Boston are expensive because people want to live there, and they also are heavily regulated. Resolving and isolating the supply-side regulatory effect from the demand-side price pressure that also independently drives regulatory stringency in many cities, through precise quantification is the central empirical challenge of our paper. In order to extricate this effect, if it exists, we specify the progressive addition of controls to build out the rest of the analysis section.

Having established the raw baseline regression, we next introduce a series of comprehensive demand-side controls that are intended to isolate the supply-side regulatory effect from the demand-driven variation in housing costs. Specifically, we elected to control for the population density of the metropolitan area, household income, the share of residents holding a bachelor's degree or higher, the population growth rate, and the decade of construction as displayed above.

As can be observed in Table 1, the WRLURI still displays a robust and statistically significant positive correlation with home values after the introduction of demand-side controls, though the coefficient declines substantially from the raw bivariate regressions. The coefficient that we observed on WRLURI implies that a one standard deviation increase in regulatory restrictiveness is associated with a home value increase of $139,816, when holding income, population density & growth, and the proportion of college educated adults constant.

To no one’s surprise, not even our own, it appears to be the case that a meaningful share of the raw correlation between the WRLURI and housing prices reflects demand-driven confounding rather than a regulatory, or supply-driven, effect. The variables corresponding to population density, household income, and college education, are all also robustly positively associated with home values. This is consistent with a standard economic assumption that housing prices, like any other commodity, are driven, on the demand side, primarily by the desirability of the attributes of the commodity. In the case of houses, the price increase that we observe is driven by the desirability of the location and the income of its residents.

Although, rather to the contrary of this, as things cannot always be so tidy, one interesting finding that we observed was that there existed a negative correlation between the employment rate and housing values, meaning that a higher employment rate in an MSA was associated with lower home values, conditional on other controls. This is a mildly surprising finding and it is unclear exactly why this association was observed. We flag this finding as an area for future investigation.

We next introduce state level fixed effects into our regression specification, absorbing all unobserved time-invariant variance at the state level, primarily with the goal of extricating the effect of state-specific legal statutes governing land use, and local economic factors that lie upstream of housing development, as well as (and including) political economy factors. It should be noted that the institutional structure of zoning law in America is derived from state authority through the police powers enumerated to state legislatures under the Tenth Amendment to the Constitution. Each individual state passes legislation that authorizes and enables local governments to enact zoning ordinances, and, owing to this, we hypothesize that there likely exist inter-state differences in the stringency and traits of local zoning regulation.

We build out the regression specification described prior again, regressing housing prices on the WRLURI 2018, our smattering array of demand side controls, and our unit quality control specification. This time, we specifically control for state level fixed effects.

The introduction of state fixed effects strongly attenuates the coefficient on our primary variable of interest, the WRLURI, as expected. A joint F-test on the state fixed

effects strongly rejects the null hypothesis that all state dummies are jointly equal to zero (F(43,15078967) = 4.56, p < 0.001), confirming that inter-state heterogeneity is a meaningful source of variation in housing costs and that state fixed effects are warranted in our specification.

The diminution of the WRLURI coefficient under the regression buildout incorporating the state fixed effects regression specification does somewhat strongly suggest that a substantial share of the cross-sectional relationship between regulatory stringency and housing prices emanates from *inter*-state legal variation rather than *intra*-state local jurisdictional legal variation. Thus, our hypothesis that state-level housing policy frameworks *in their own right*, not merely local zoning ordinances, their procedures and desiderata, are an important determinant of housing prices and subsequently housing affordability appears to preliminarily be supported by the evidence. We return to this finding in the discussion section.

Table 2: OLS Estimates: Price-to-Income Ratio

	(1)	(2)	(3)
VARIABLES	Baseline	Controls	Controls + State FE
WRLURI (2018)	17,270***	12,164***	5,916***
	(13.37)	(17.05)	(21.19)
Population Density		0.526***	0.524***
		(0.00276)	(0.00285)
Household Income		0.104***	0.0990***
		(0.000178)	(0.000176)
College Attainment Rate		6,321***	6,053***
		(19.31)	(18.77)
Employment Rate		-124,885***	331,390***
		(799.3)	(1,286)
Population Growth Rate		1,294***	-6,825***
		(74.28)	(110.9)
Constant	28,125***	127,807***	-303,247***

	(5.968)	(751.4)	(1,202)
Observations	30,350,865	14,989,080	14,989,080
R-squared	0.062	0.220	0.264
R-squared	0.0618	0.220	0.264

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

We next replicate our full regression sequence again, under these revised conditions electing to employ the price-to-income ratio as our primary dependent variable. The price-to-income ratio (PTI), which we define as the home value divided by the log of household income is our preferred measure of housing affordability as it standardizes housing costs relative to the purchasing power of local residents, making it comparable across metropolitan areas of very different income levels and more directly relevant to the policy question of housing access.

The PTI ratio results largely corroborate the home value findings but with some important differences. The WRLURI coefficient in the bivariate PTI specification is positive and significant, confirming that more restrictive regulatory environments are associated with decreased affordability even after normalizing for income levels. This is an important result. It implies that the regulatory effect on housing costs is not simply a reflection of the fact that high-income areas can afford higher prices, but rather that regulatory restrictiveness independently raises the cost of housing relative to the incomes of local residents.

However, we note that the mechanical relationship between PTI ratio and income complicates the interpretation of income as a control variable in the PTI regressions, as income appears in the denominator of the dependent variable and obviously as an independent variable in our default control specification for our regression.We therefore

report the PTI results both with and without income as a control in Table 3, and note that our preferred specification uses population growth, bachelor's degree share, and employment rate as demand-side controls in place of income when PTI is the dependent variable. The WRLURI coefficient is robust across both specifications.

Subgroup Analysis:

Having established the baseline relationship between the aggregate WRLURI and housing costs, we turn to our primary contribution: decomposing and estimating the effect of regulatory stringency on housing prices across the regulation-type specific sub-indices of the WRLURI in order to identify which dimensions of regulatory restrictiveness most strongly predict housing costs. We centered our analysis on three specific sub-indices of primary theoretical interest. Number one, the Density Restriction Index (DRI), which attempts to capture the range of variation in minimum lot requirement regulations. Number two, the Approval Delay Index (ADI), which attempts to quantify the extent to which the review and approval process for a new erection varies between municipalities – as this is one of the main roadblocks to new construction. And last but not least, Number 3, the Supply Restriction Index (SRI) which reflects the extent to which there exist constraints or caps on the supply of new homes to the market.

Table 2A: Sub-Index Regressions on Home Value

	(1)	(2)	(3)	(4)	(5)	(6)
VARS	DRI Only	DRI + Unit	DRI + All Controls	DRI + State FE	ADI Only	ADI + Unit
DRI18	-28,587***	-251,467***	-180,582***	-44,568***		
	(998)	(787)	(1,147)	(1,636)		
ADI18					118,502***	3,948***

					(219)	(177)
Rooms		-243,987***	-201,058***			-247,764***
		(249)	(366)			(250)
Pop. Density			-1*** (0)			
HH Income			-2*** (0)			
College			-418,706*** (1,515)			
Emp. Rate			-14,659,690*** (74,053)			
Pop. Growth			2,169,508*** (7,684)			
Obser-vations	44,381,008	42,623,401	21,199,419	21,199,419	44,278,806	42,523,642
R^2	0	0	0	0	0	0
Unit Control	No	Yes	Yes	Yes	No	Yes
Demand Control	No	No	Yes	Yes	No	No
State FE	No	No	No	Yes	No	No

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

We first regress home value on each of our chosen sub-indices individually, including our full demand-side control vector and state fixed effects. The results reveal substantial heterogeneity across regulatory dimensions. The ADI initially displays the strongest and most robust positive association with home values among all sub-indices, with a coefficient of approximately $118502 per unit increase in approval delay index score, significant at the 1% level ($p < 0.01$). This finding is intuitive: longer permitting timelines directly impede the construction of new housing supply by increasing carrying costs, financing uncertainty, and effective development costs, generating a supply constraint that manifests in elevated prices.

However, when a full regression buildout is specified, the SRI, in turn, displays the strongest and most robust positive association with home values, clearly evincing the pronounced negative effects of explicit annual caps on building permits and construction activity on prices. This effect is mediated directly through its role in constraining the ability of housing suppliers to meet demand.

The DRI, by contrast, displays a negative coefficient in the hedonic regression specification – Communities with more stringent minimum lot size requirements are associated with lower, instead of higher, home values. This initially counterintuitive finding likely reflects a composition effect: densely developed urban areas, which tend to have the highest home values on a per-unit basis, often have lower formal density restrictions than low-density suburban and exurban communities where large minimum lot size requirements are most prevalent.

If possible, we would control for this by using price per square foot rather than total home value as the dependent variable, however, the ACS microdata that we have access to has the limitation of not reporting square footage per home. This is an unfortunate and limiting constraint for our study. Nonetheless, the coefficient on DRI being negative initially is a very important finding and directly contradicts our hypothesis.

However, in a surprising twist of fate, when we switch to the preferred price-to-income ratio measure that we specify in our methodology section, we see a significant sign flip on the DRI coefficient, where increases in the degree of density restriction – communities with bigger minimum lot size requirements, reducing the

degree of allowable density– are indeed significantly associated with an increase in the price to income ratio in that jurisdiction, and thus, a decrease in the affordability of housing.

Table 2A (Cont'd): Sub-Index Regressions on Home Value

VARIABLES	(7) ADI + All Controls	(8) ADI + State FE	(9) SRI Only	(10) SRI + Unit	(11) SRI + All Controls	(12) SRI + State FE
	(74,264)			(52,865)	(74,244)	
Rooms	-201,576***			-247,454***	-201,591***	
	(367)			(249)	(366)	
Density	-4***				-2***	
	(0)				(0)	
HH Income	-2***				-2***	
	(0)				(0)	
College	-427,405***				-420,340***	
	(1,517)				(1,514)	
Emp Rate	-16,070,040***				-16,298,839***	
	(73,940)				(73,602)	
Population Growth	2,523,588***				2,405,351***	
	(7,564)				(7,514)	
ADI18	35,416***	-17,653***				
	(260)	(433)				
SRI18			86,822***	118,499***	136,106***	56,809***
			(1,706)	(1,393)	(1,968)	(2,124)
Observations	21,154,171	21,154,171	44,462,036	42,700,365	21,238,026	21,238,026
R-squared	0	0	0	0	0	0
Unit Controls	Yes	Yes	No	Yes	Yes	Yes
Demand Controls	Yes	Yes	No	No	Yes	Yes
State FE	No	Yes	No	No	No	Yes

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 2B: Sub-Index Regressions on PTI Ratio

VARIABLES	(1) DRI	(2) ADI	(3) SRI	(4) Joint	(5) Joint + WRLURI
WRLURI18					2,102.487***
					(24.642)
DRI18	1,055.107***			-479.503***	-719.008***
	(15.263)			(16.134)	(17.243)
ADI18		1,767.631***		1,950.720***	1,705.716***
		(5.074)		(5.068)	(5.752)
SRI18			328.801***	-1,622.085***	-1,756.382***
			(31.380)	(31.272)	(31.562)
Observations	15,075,995	15,041,361	15,105,710	15,011,646	14,989,080
R-squared	0.286	0.292	0.286	0.293	0.293
Unit Controls	Yes	Yes	Yes	Yes	Yes
Demand Controls	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

When we include DRI, ADI, and SRI simultaneously in a single regression alongside the full control vector and state fixed effects (Table 2B), the ADI remains positive and significant while the DRI becomes more strongly negative. Confusingly, the SRI also flips signs and becomes negative. We discuss these results further in the discussion section below. The dominance of ADI in the joint specification is consistent with the finding of Molloy, Nathanson and Paciorek (2022) that the process constraints on housing supply are the binding regulatory mechanism in markets where housing demand is inelastic, rather than density or quantity constraints.

Table 2C: Joint Sub-Index Regression on Home Value

VARIABLES	(1) Joint - No Controls	(2) Joint + Unit	(3) Joint + All Controls	(4) Joint + State FE	(5) Joint + WRLURI + State FE
WRLURI18					-24,301***
					(2,504)
DRI18	-88,996***	-264,588***	-190,407***	-29,565***	-20,878***
	(1,018)	(804)	(1,171)	(1,729)	(1,777)
ADI18	124,061***	5,047***	33,375***	-20,863***	-18,073***
	(225)	(180)	(264)	(459)	(551)
SRI18	-74,546***	130,553***	106,226***	77,679***	78,865***
	(1,730)	(1,405)	(1,982)	(2,172)	(2,174)
Observations	44,197,778	42,446,678	21,115,564	21,115,564	21,083,605
R-squared	0	0	0	0	0
Unit Controls	No	Yes	Yes	Yes	Yes
Demand Controls	No	No	Yes	Yes	Yes
State FE	No	No	No	Yes	Yes

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

This result is also noteworthy from an identification standpoint. Approval delays are less susceptible to the reverse causality concern that motivates our endogeneity discussion — it is more plausible that expensive communities adopt more restrictive density limits in response to high prices than that they lengthen their permitting timelines for the same reason, since delay is more likely to be an institutional and administrative characteristic than a deliberate exclusionary policy. The robustness of the ADI result therefore provides some additional confidence that our findings reflect a real supply-side regulatory effect and not reverse causation.

Endogeneity & Causality:

A central concern remains in our cross-sectional analysis; the potential for reverse causality and endogeneity remain ever-present in our regression specifications. The concern, stated plainly, is that communities that are already expensive might adopt stricter zoning and land use regulations precisely in order to preserve their exclusive nature, and to protect the property values of incumbent homeowners. Now this does not really run counter to our thesis, but rather simply shifts the perspective from a matter of bureaucratic legal procedures unintentionally constraining the ability of supply to meet demand, *thus driving prices higher,* to the deliberate political economy of exclusionary zoning being used to protect existingly high price markets. While the laws still constrain supply in the latter case, they in theory may not serve much to drive prices higher, rather merely keeping them high.

We devised a rudimentary test to reject the reverse causality theory in its more extreme cases. If expensive communities systematically adopt stricter regulations in response to higher prices, then 2006 prices would be a significant predictor of the change in WRLURI scores between 2006 and 2018. We found that this was not the case. The coefficient on 2006 home values in a regression of the WRLURI delta between the 2006 and 2018 editions on pre-period prices and controls was negative, although not particularly convincing. That is, higher prices predicted deregulation, which suggests that the reverse causality mechanism is not operative here. Maybe in some jurisdictions, some small portion of the degree of regulatory tightening may in theory still be endogenous to house prices, but the vast majority of regulatory variation that we observe should not be attributable to it.

We also tested whether or not the 2006 WRLURI scores could predict 2018 housing costs, controlled for 2006 prices. In theory, if reverse causality were indeed the case, the regulatory intensity should lose its predictive power if one adequately controls pre-period prices, because the regulation would mostly be reflecting those prices, more so than it constrains supply. We found some evidence of the 2006 WRLURI predicting 2018 housing prices with a high degree of statistical significance, controlling for 2006 home values. This is consistent with our view that the regulation itself is the independent, main driver of housing prices itself.

Table E1: Endogeneity and Reverse Causality Tests

VARIABLES	(1) 2018 HV on 2006 WRLURI (C 2006 HV)	(2) ΔWRLURI on 2006 HV	(2) ΔWRLURI on 2006 PTI
WRLURI06	36,918***		
	(1,385)		
2006 HV	0***	-0.000***	
	(0)	(0.000)	0.000***
PTI ratio 2006			(0.000)
Observations	448,260	11,984,172	11,984,172
R-squared	0	0.529	0.529
Demand Controls	Yes	Yes	Yes
State FE	Yes	Yes	Yes

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Discussion:

The results of our statistical regression produced numerous substantive findings, that, taken together, cleanly lay out the nature of the relationship between local land use

regulation and housing affordability in American cities. First of all, and perhaps most importantly, we find robust evidence of a statistically significant positive relationship between our preferred measurement of regulatory intensity, the aggregate WRLURI, and housing costs across our full sample of metropolitan areas. This relationship is so strong, that it persists and is consistently observed no matter what affordability metric we use, and survives the introduction of our slew of comprehensive demand side controls and unit quality measures. This finding is in line with the existing literature examining the relationship between housing prices and regulatory stringency, especially those studies that instrument the WRLURI.

The regression coefficient on the WRLURI does decline substantially, from a $206,938 to $139,816 increase in home value per standard deviation of regulatory restrictiveness, upon the introduction of these controls into the regression model, however, this is 100% in line with any economically informed expectation, and it is highly informative. It confirms the assumption that a large proportion of the regulatory-price correlation is a byproduct of demand-driven price increase, yet there still remains more than substantial evidence of a residual regulatory effect even after accounting for a full slew of demand-side factors.

Moreover, we find that the relationship is significantly attenuated to a further extent–but critically not eliminated–when we introduce state-level fixed effects into the regression equation. The coefficient on WRLURI declines to a $68,614 increase in home value per standard deviation of increase in regulatory restrictiveness. We ran a test to verify the significance of the state fixed effects and were able to confirm their joint significance. The F-Statistic was 4.56 on 43 degrees of freedom, significant at the 1%

level, and supports the notion that inter-state legal variation is also a meaningful predictor of cross-metropolitan housing cost differences. We interpret this as evidence that state-level housing frameworks, in further addition to local zoning ordinances, are also an independent predictor of housing affordability outcomes. State fixed effects were a great addition to the regression specification.

There are obvious policy implications to this finding. It is clear then, that reforms targeting state-level housing policy and enabling legislation may be sufficient alone to reduce housing costs in the absence of local regulatory change. And, to the corollary, reforms that target local zoning alone may be *in*sufficient to effectively address affordability issues if the binding constraint on housing supply is primarily downstream of state-level zoning legislation and regulatory frameworks.

For the final and perhaps most novel finding, we find that our WRLURI sub-index decomposition reveals substantial heterogeneity in which regulatory mechanisms affect housing affordability measures the most, and to what extent. We consistently observe that the Approval Delay Index of the WRLURI emerges as the top predictor of housing costs across our buildout of specifications. The coefficient on the ADI is consistently highly positive and highly significant, with a great degree of robustness. Conversely, the DRI displayed a highly counterintuitive negative coefficient in our naive hedonic regressions, but reversed signs when we substituted the price to income ratio in for the dependent variable, as our preferred affordability metric.

Approval Delay Index:

The persistent performance of the ADI across our sub-index decomposition regressions represents the most substantial finding in our analysis. In the individual sub-index specifications, the ADI produced a coefficient representing a $118,502 increase in home value per unit increase in the approval delay score in the naive hedonic regression, and a highly significant increase in the price to income ratio per unit increase in the approval delay score. The ADI consistently displayed the largest and most robust coefficients among our three focal sub-indices, and this effect was observed consistently across our control specifications. Even in the joint PTI specification, when DRI and SRI were included in the regression alongside it, the ADI coefficient remained positive and highly significant. This substantiates the notion that the approval delay effect is independent of the density and supply restriction effects.

This finding is consistent with the theoretical predictions of models in which permitting timelines function as a tax on new housing construction. Longer approval processes increase the carrying costs of development — the interest on construction financing, the opportunity cost of land held in pre-development limbo, and the uncertainty premium demanded by developers operating in environments with unpredictable regulatory timelines. These costs are ultimately passed through to housing consumers in the form of higher prices, reducing the affordability of new construction and limiting the supply response to demand shocks.

The implications of the ADI finding for public policy are straightforward. Our ADI finding strongly suggests that streamlining the approval process for new residential

construction will be a great addition to any pro-housing policy suite being considered by lawmakers. Moreover, passing a bill that fast-tracks permitting and approval requires minimal time spent in the regulatory weeds by lawmakers, carefully pruning and manipulating highly technical building regulations and navigating intensely politically contentious changes to zoning codes. Lawmakers can instead effect these changes through staffing increases, statutory timelines, and lessening (or removing) mandated delays, or even through more politically explosive means, like pursuing by-right zoning policies. Thus, it is likely one of the highest-leverage interventions available for those who seek to improve housing affordability.

DRI Sign Reversal:

The negative coefficient on the DRI in our naive hedonic regression initially appears to contradict our hypothesis that an increase in density restrictions would increase housing prices and decrease housing affordability. It suggests, superficially at least, that communities with more stringent minimum lot size requirements, and thus more density restrictions, have reduced home values. However, we posit that this finding is confounded by substantive differences in the composition of neighborhoods with high density restrictions, and that the DRI coefficient is rather capturing the suburban/urban split, rather than a truly negative causal relationship between density restrictions and housing affordability.

Densely developed urban communities, as powerhouses of industry and economic might, concentrate the most people and the most jobs in one place, and hence possess the highest per-unit home values, on account of exceedingly high

demand. And because these cities are so built out and densely populated, they cannot in practice have formal density restrictions that are as stringent as those of low-density suburban and exurban communities where large minimum lot size requirements are most prevalent. Otherwise, it would have the effect of de-facto banning development altogether–such as has effectively occurred in San Francisco. A Manhattan condo is extraordinarily valuable because it is located in Manhattan, an economic powerhouse that generates millions of high paying, high value output jobs, while a half-acre lot in rural Michigan displays comparatively modest home values, because the underlying land and structure are worth less by orders of magnitude. Thus, the existing negative relationship between density restrictions and real estate values in the raw data results in the negative DRI coefficient we observe.

Yet, when we switch to our preferred affordability measure, the price-to-income ratio, and set it as our dependent variable, the DRI coefficient reverses signs and displays a significant positive association with decreased housing affordability. In other words, communities with more stringent minimum lot size requirements are associated with less affordable housing. In support of our argument, this phenomenon is precisely what one would expect to find if it were indeed true that density restrictions operate primarily by constraining the supply of housing relative to local demand. Thus, prices are raised relative to incomes, even so in markets where the absolute home values are orders of magnitude lower than in densely populated urban areas.

The lack of a square footage variable in our ACS dataset, subsequently leaves us unable to control for the price per unit of square footage. Thus, we are unable to directly control for unit size and test whether density restrictions raise the relative cost of

housing. This is an unfortunate limitation of our analysis, and makes it problematic to declare confidently that the DRI sign flip is solely a matter of compositional confounding . We flag this as an important topic for future research.

SRI Sign Reversal:

Another surprising finding that warrants discussion in this section is the observed sign reversal on the SRI coefficient when we included it in a joint regression alongside the DRI and ADI. In the individual regression specification, the SRI displayed a positive and statistically significant correlation with our measures of housing affordability, as we expected. This is consistent with the notion that explicit annual caps on housing construction constrain supply and raise prices. However, when we included it in a joint regression, with DRI and SRI, on the PTI ratio, the SRI and DRI coefficients became strongly negative, with the ADI absorbing all of the positive signal.

We charitably interpret this sign reversal primarily as an artifact of multicollinearity between the WRLURI subindices rather than a substantial finding. It is not terribly surprising that the three variables would be correlated with each other, in light of the fact that they are all measuring direct supply-side constraints on housing, taken from the same index. Communities likely have a great degree of homogeneity among the sub-indices of the WRLURI. That is, highly regulated communities are much more likely to be relatively highly regulated in all of their dimensions. If they score high on one dimension, they score high on others.

As a result of this, these sub-indices collectively share redundant information, making it difficult to disentangle which coefficients are contributing what. As a result,

coefficients become unstable and may flip signs, even though the joint predictive power is clearly identified and preserved. Our interpretation is consistent with the fact that coefficient on the SRI is positive and statistically significant in all individual regressions and only becomes negative under the joint regression specification. Regardless, the ADI retains the largest and most stable positive coefficient across all specifications, which speaks to the importance of its contribution to increasing housing costs.

State Fixed Effects

That the coefficient on the WRLURI variable was significantly attenuated, decreasing from $139,816 to $68,614 under the home value regression specification, and from $12,164 to $5,916 under the PTI ratio regression specification is particularly noteworthy. This attenuation primarily reflects two distinct potential phenomena.

The most optimistic interpretation given our priors in this study is that inter-state variation in regulatory restrictiveness is a major channel through which regulation affects housing affordability. State level housing policies, on top of local zoning ordinances, have a significant effect on housing prices and act as a binding restraint on housing supply in practice, or exert a significant chilling effect on new homes being supplied to market. This interpretation supposes that the state fixed effects are absorbing regulatory variation and that the coefficient corresponding to the WRLURI variable, post-decrease, mostly represents the effect of local regulatory restrictiveness net of the state-level institutional effect.

Yet another interpretation of this observed coefficient decrease is that the state fixed effects are merely absorbing some other unobserved confounding variables that

happen to vary at the state level, yet are also correlated with the degree of local regulatory stringency and housing costs. It is not exactly clear what this could be, although it is certainly possible that it is simply picking up on demand driven variation. High-cost states like California, New York, or Massachusetts are subject to intense demand-side pressure because of their spectacular quality of life and public amenities (healthcare, transportation, labor laws, etc) and they also happen to be heavily regulated, especially in the context of housing. Perhaps the decrease in the WRLURI coefficient under state fixed effects is partly, or mostly, attributable to absorption of some sort of residual demand-side confounding, rather than a reduction in the degree to which regulatory stringency affects housing costs.

We are not able to distinguish between the numerous explanations for the significance of the state fixed effects result, although it is well beyond the scope of our paper to do so. This remains an area for future research and analysis. Speculatively, we suggest that in light of the fact that the coefficient on the WRLURI variable remains positive, the fixed events are also statistically significant, it seems probable that multiple interpretations are equally at play here, each one responsible for a share of the reduction.

In the future, disentangling the effect of demand-driven residual confounding from direct state regulatory policy per-se is a promising area for future research that has important policy implications in the ongoing debate over housing reform.

Robustness & Limitations of our Design

We are very aware of the potential for endogeneity, and also the potential for reverse causality in our analysis. It could be the case, for example, that communities that are already expensive enact stricter regulations in order to preserve exclusivity.

Although we do not suppose that this is necessarily at odds with the logic of our hypothesis–in some cases, zoning regulations are enacted specifically with the intent of keeping house prices high–we acknowledge that reverse causation nevertheless remains a lingering possibility. We also recognize that there are a number of limitations that are inherent in our research design. First of all, the cross-sectional nature of our core analysis does frustrate our ability to make causal claims. Though we thoroughly attempted to employ a difference-in-difference, quasi-experimental study design, comparing the change in outcomes between treated metros (high WRLURI delta between 2006 and 2018) vs untreated metros (low WRLURI delta), we ultimately did not have enough statistical power to perform our analysis. And, in light of crippling technical merge issues, we were left unable to generate more. Furthermore, although we attempt to control for a smattering array of variables in our analysis, we cannot possibly foresee and preempt all possible sources of unobserved residual confounding.

Finally, the WRLURI, while certainly a remarkably simple and utilitarian study tool, is effectively no more than two snapshots in time. It captures the level of restrictiveness of a local housing regulatory regime in a given place at a given time, and gives us information on how this restrictiveness differs across multiple dimensions of regulatory constraint, but does not provide any data on the pace of change. It is not a time series dataset. Subsequently, our analysis is limited purely to the correlation

between the long-run regulatory equilibrium and the price level, rather than the short run effects of regulation or deregulation. We do not wish to overstate the causal certainty of our findings in this analysis.

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